Experiment 3: Data Visualization and Correlation Analysis

Aim

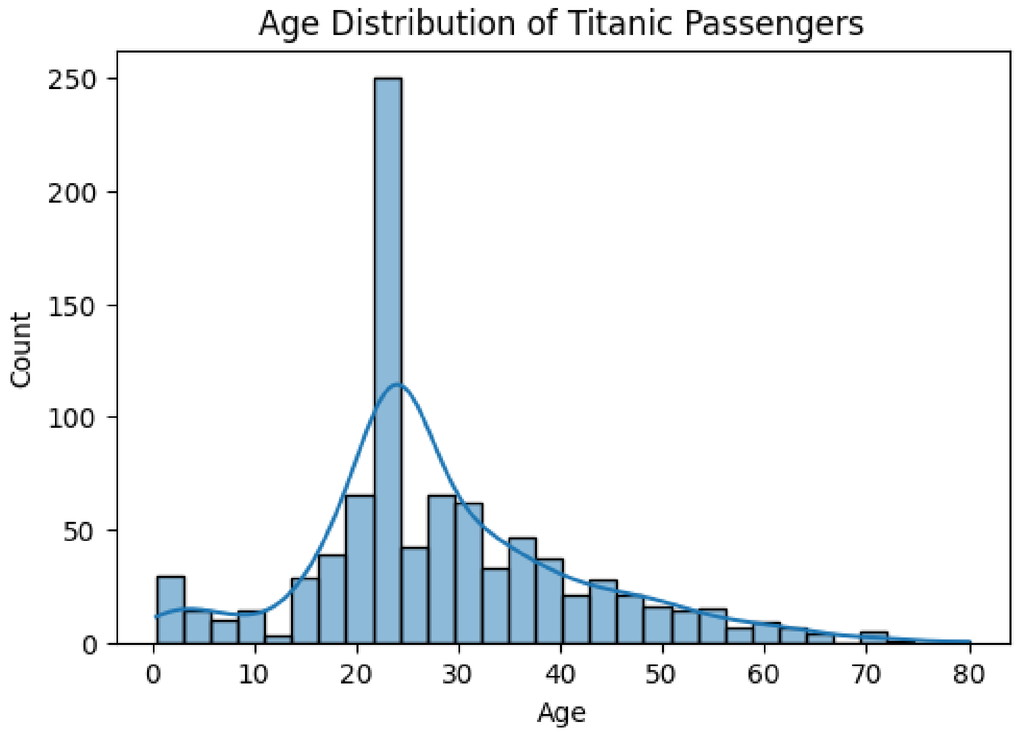
To perform data preprocessing and visualize the dataset using various plots such as histograms, count plots, bar plots, scatter plots, box plots, and heatmaps to understand data distribution, relationships, and correlations.

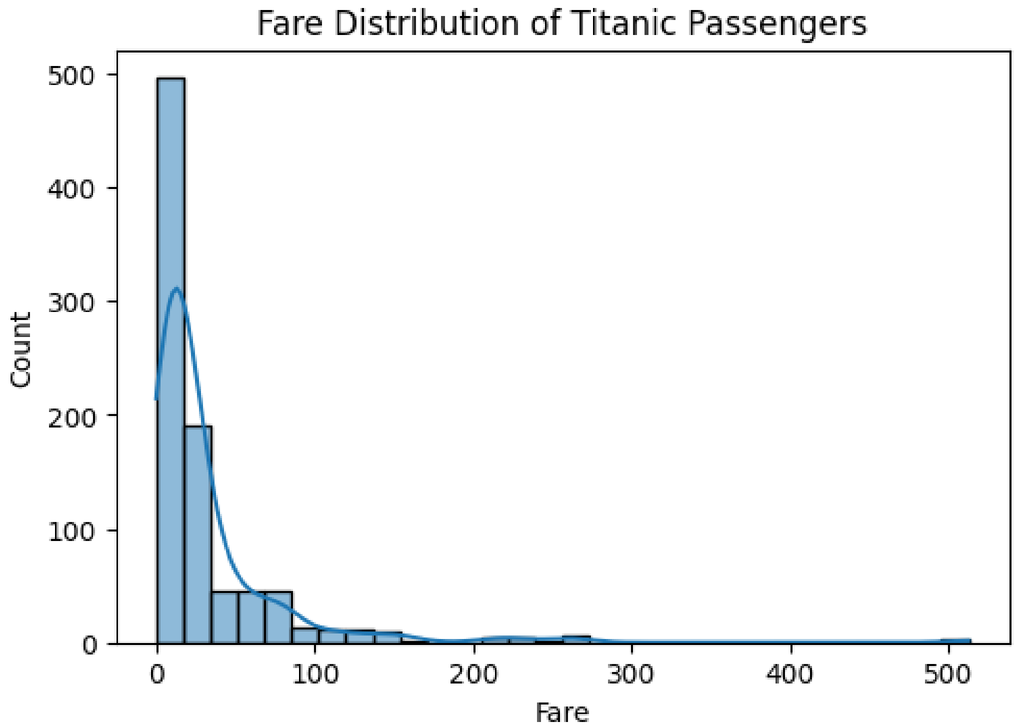
Code

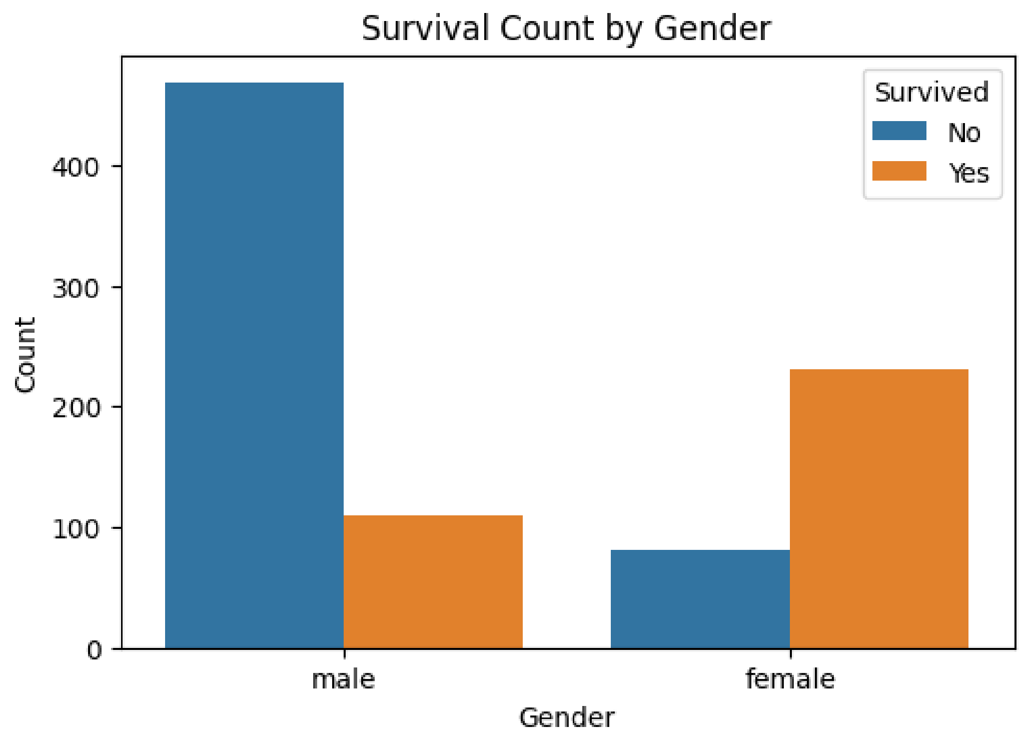
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset('titanic')
df.head()

df['age'].fillna(df['age'].mode()[0],inplace=True)
df.drop(columns=['deck'],inplace=True)
df.dropna(inplace=True)
df.isnull().sum()

plt.figure(figsize=(6,4))
sns.histplot(df["age"], bins=30, kde=True)
plt.title("Age Distribution of Titanic Passengers")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()

plt.figure(figsize=(6,4))
sns.histplot(df["fare"], bins=30, kde=True)
plt.title("Fare Distribution of Titanic Passengers")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()

plt.figure(figsize=(6,4))
sns.countplot(data=df, x="sex", hue="survived")
plt.title("Survival Count by Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()

plt.figure(figsize=(6,4))
sns.barplot(data=df, x="pclass", y="survived")
plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
```

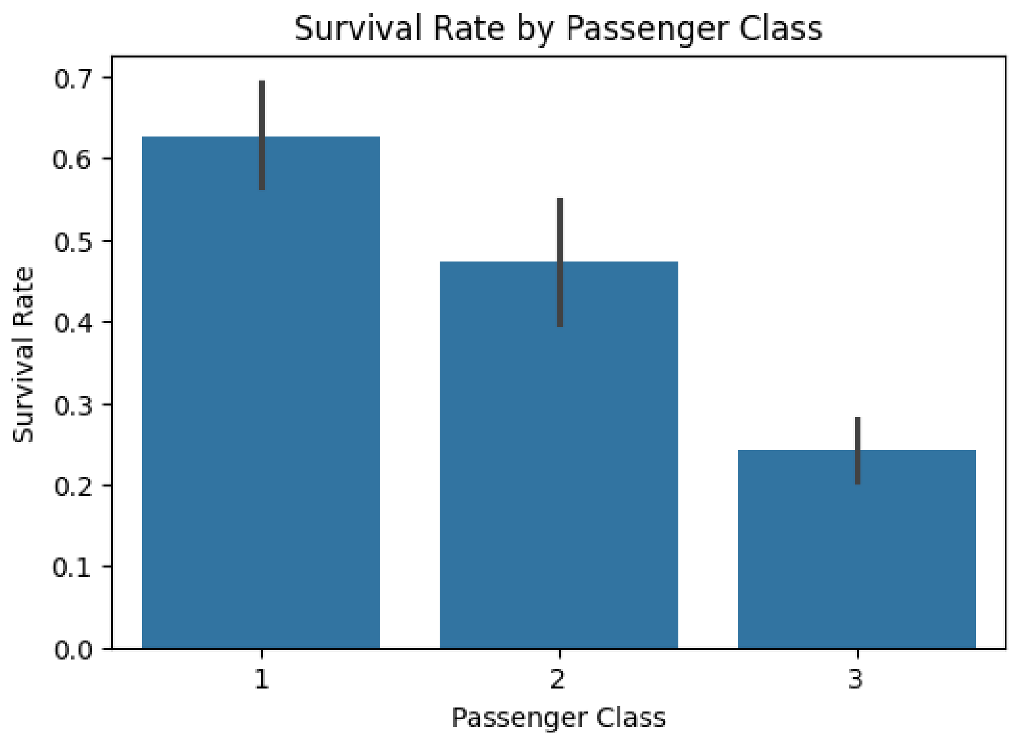
```
plt.ylabel("Survival Rate")
plt.show()

plt.figure(figsize=(6,4))
sns.scatterplot(
    data=df,
    x="age",
    y="fare",
    hue="survived",
)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(data=df, x="pclass", y="fare")
plt.title("Fare Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Fare")
plt.show()

numeric_df = df.select_dtypes(include=['int64', 'float64'])

corr = numeric_df.corr()

plt.figure(figsize=(8,6))
sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

plt.title("Correlation Heatmap of Titanic Dataset")
plt.show()
```

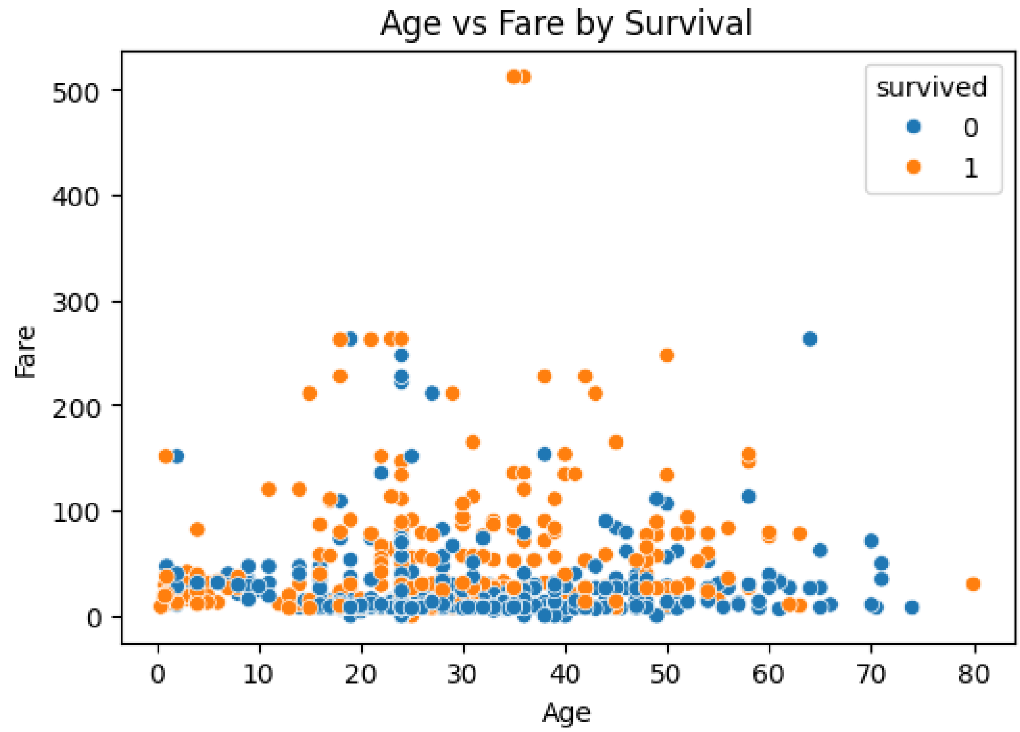
Output


Age Distribution of Titanic Passengers
Count
Age
0
50
100
150
200
250
0
10
20
30
40
50
60
70
80

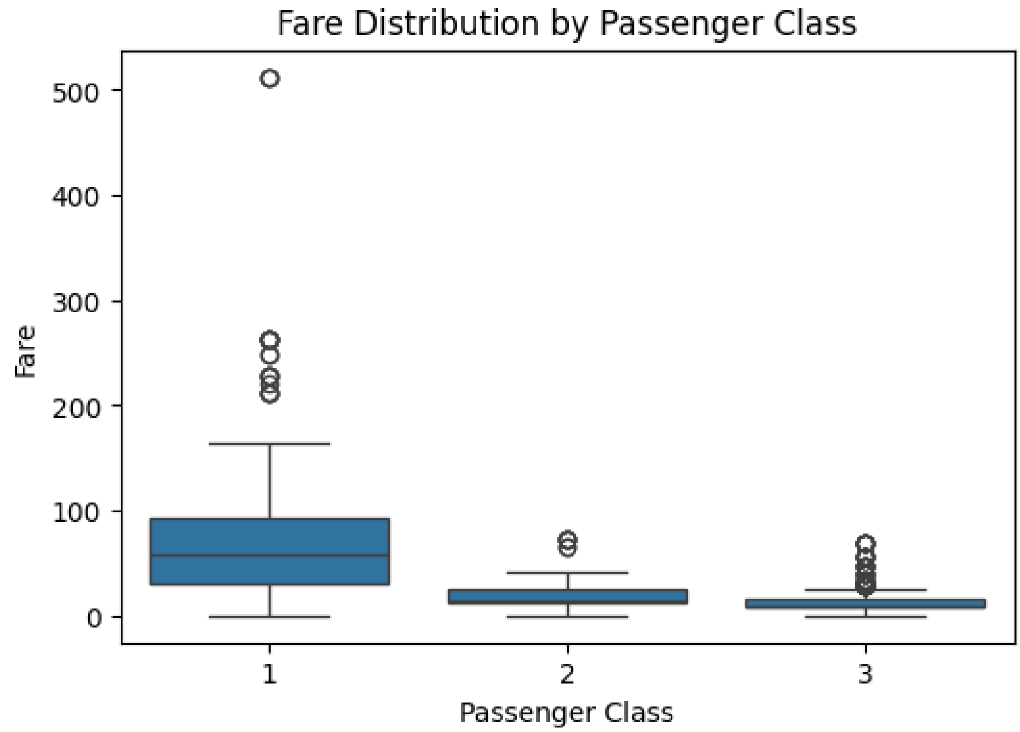

Fare Distribution of Titanic Passengers
Count
Fare
0
100
200
300
400
500
0
100
200
300
400
500


Survival Count by Gender
Survived
No
Yes
Count
400
300
200
100
0
male
female
Gender

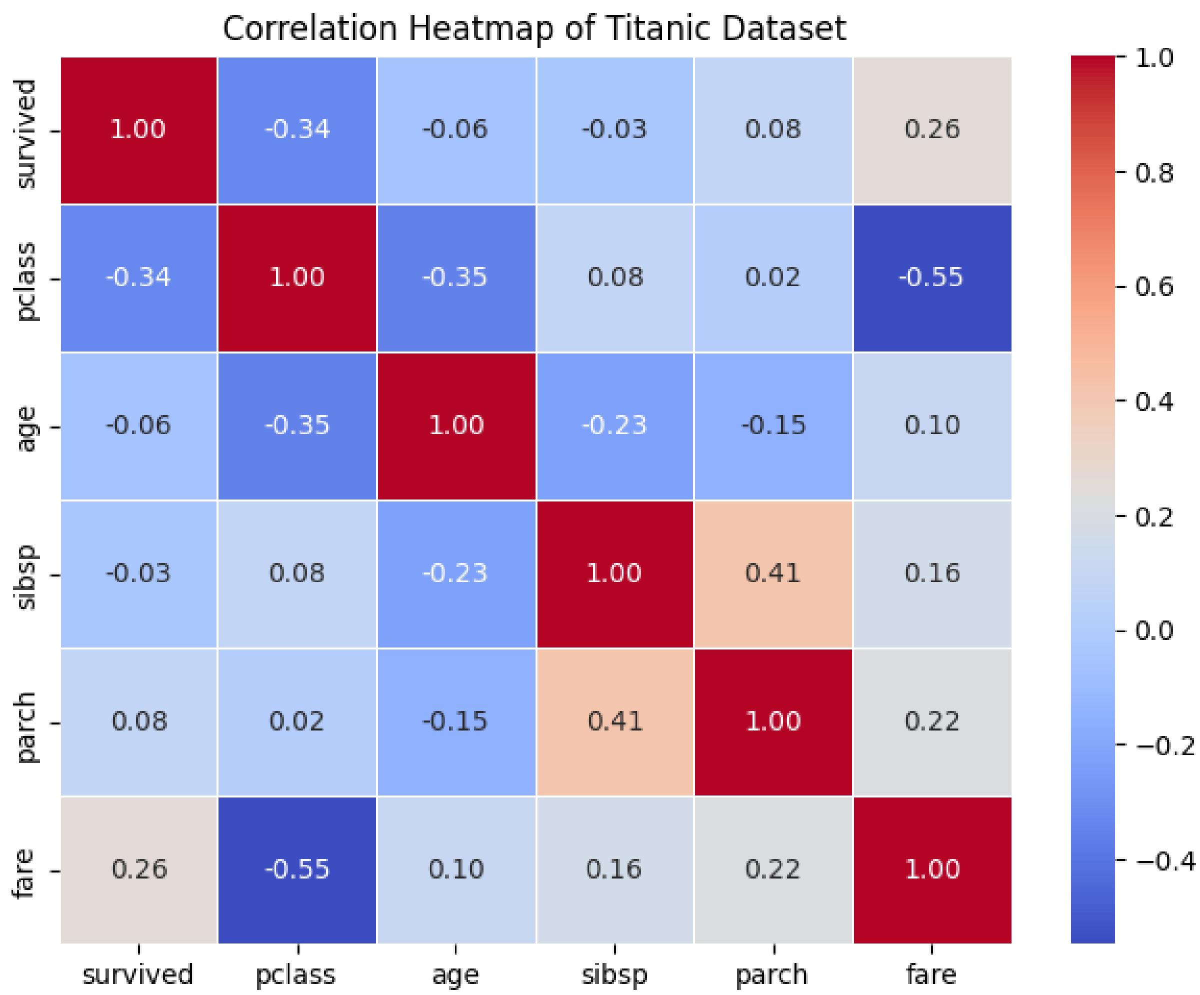

Survival Rate by Passenger Class
Survival Rate
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
1
2
3
Passenger Class



Age vs Fare by Survival
survived
0
1
Fare
Age
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80



Fare Distribution by Passenger Class
Fare
Passenger Class
0
100
200
300
400
500
1
2
3



Result

The dataset was successfully preprocessed by handling missing values and removing unnecessary columns. Various visualizations were generated to analyze the data distribution and relationships between variables. Histograms showed the distribution of age and fare, count and bar plots revealed survival patterns based on gender and passenger class, while scatter and box plots illustrated relationships and variations in data. The correlation heatmap provided insights into the strength and direction of relationships between numerical features, helping in better understanding of the dataset.